Departement Elektriese, Elektroniese en Rekenaar-Ingenieurswese

Department of Electrical, Electronic and Computer Engineering

Eksamen
Kopiereg voorbehou

Examination
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Module EBN111
Elektrisiteit en Elektronika
18 Junie 2013

Module EBN111
Electricity and Electronics
18 June 2013

UNIVERSITEIT VAN PRETORIA
UNIVERSITY OF PRETORIA
YUNIBESITHI YA PRETORIA

Eksamensinligting: ***Exam information:***			
Maksimum punte: *Maximum marks:*	90	Volpunte: *Full marks:*	90
Duur van vraestel: *Duration of paper:*	180 minute *180 minutes*	Oopboek / toeboek: *Open / closed book:*	Toe *Closed*
Totale aantal bladsye (hierdie blad ingesluit): *Total numer of pages (including this page):*		30	

BELANGRIK- ***IMPORTANT***

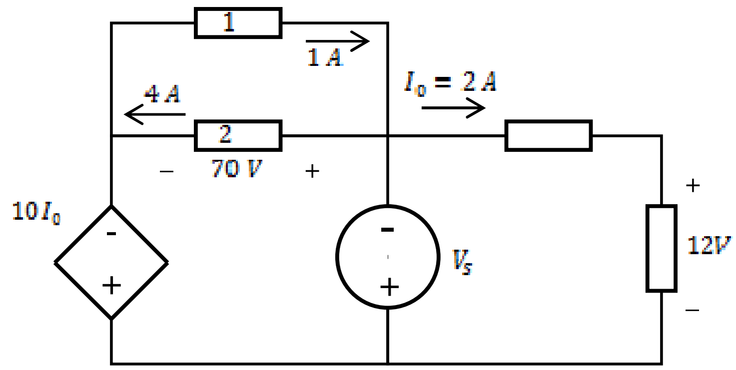
1. Die eksamenregulasies van die Universiteit van Pretoria geld.
   The examination regulations of the University of Pretoria apply.
2. Alle vrae moet op die ANTWOORDBLAD beantwoord word.
   All questions must be answered on the ANSWER SHEET.
3. Studente mag hul eie rowwe berekeninge op die vraestel doen – die vraestel word nie ingeneem nie.
   Students may do their own rough calculations on the question paper – the question paper will not be taken in.
4. Antwoorde moet eenhede insluit!
   Answers must include units!
5. Polêre vorm: $A\angle\theta$; Reghoekige vorm: $a + jb$
   Polar form: $A\angle\theta$ *; Rectangular form:* $a + jb$
6. **Finale antwoorde moet as reële getalle geskryf word, en nie as breuke nie.**
   Final answers must be written as real numbers, and not as fractions.

Dosent(e): *Lecturer(s):*	J. Joubert J.D. le Roux
Eksterne eksaminator: External examinor:	J.W. Odendaal

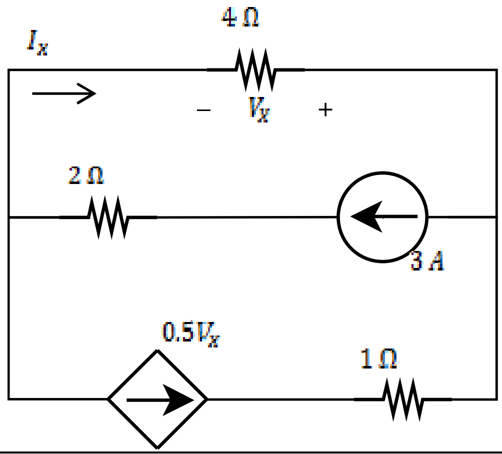
VRAAG/QUESTION 1 [18]

Vraag 1.1 [3]	**Question 1.1 [3]**
\`n 12 Volt battery voorsien 43.2 kJ oor 2 ure aan \`n gloeilamp. a) Wat is die stroom deur die gloeilamp? b) Wat is die hoeveelheid lading wat deur die gloeilamp vloei? c) Indien elektrisiteit R1/Wh kos, wat is die koste om die gloeilamp vir 2 ure aan te hou?	A 12 Volt battery supplies 43.2 kJ over 2 hours to a lightbulb. a) What is the current through the lightbulb? b) What is the amount of charge flowing through the lightbulb? c) If electricity costs R1/Wh, what is the total cost to keep the lightbulb on for 2 hours?

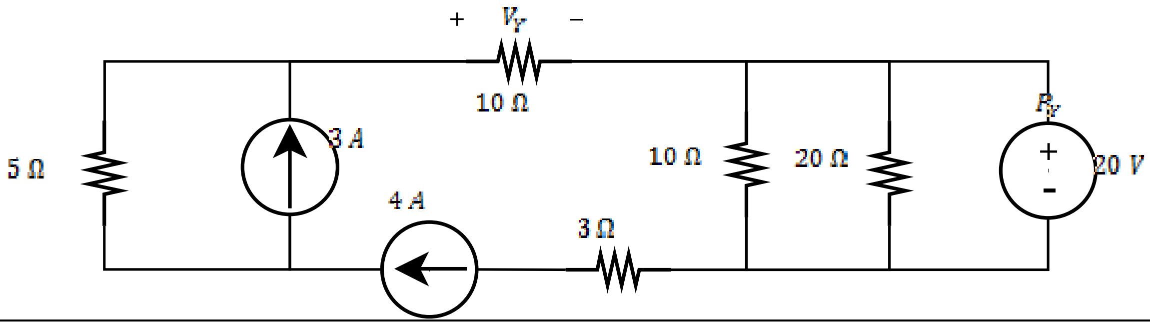
Vraag 1.2 [3]	Question 1.2 [3]
a) Hoeveel drywing absorbeer die afhanklike bron? b) Bepaal V_S indien die drywing geassosieer met die onafhanklike bron – 250 W is. c) Wat is die som van die totale drywing geabsorbeer deur elemente 1 en 2?	a) What is the power absorbed by the dependent source? b) Determine V_S if the power associated with the independent source is – 250 W. c) What is the sum of the total power absorbed by elements 1 and 2?



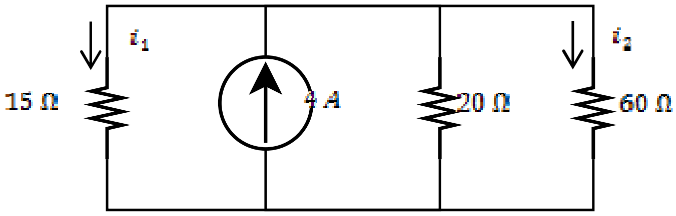
Vraag 1.3 [2] Bepaal V_X en I_X.	**Question 1.3 [2]** Determine V_X and I_X.



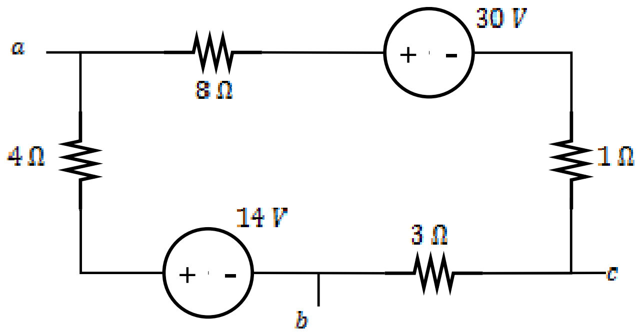
Vraag 1.4 [2] Bepaal V_Y en die drywing P_Y geassosieer met die 20 V spanningsbron.	**Question 1.4 [2]** Determine V_Y and the power P_Y associated with the 20 V voltage source.

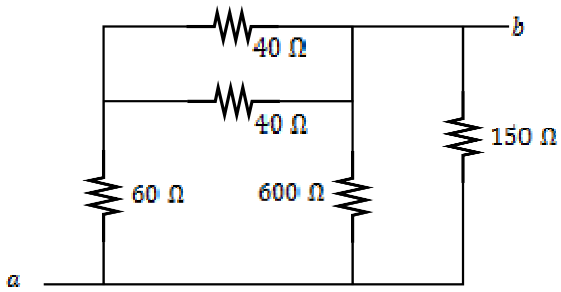
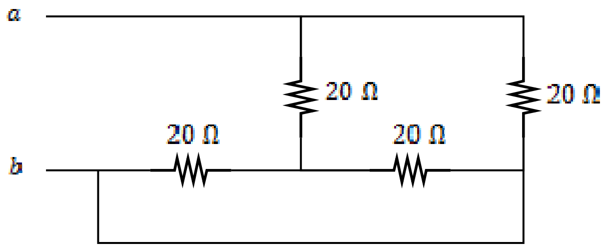


Vraag 1.5 [2] Bepaal i_1 en i_2.	**Question 1.5 [2]** Determine i_1 and i_2.



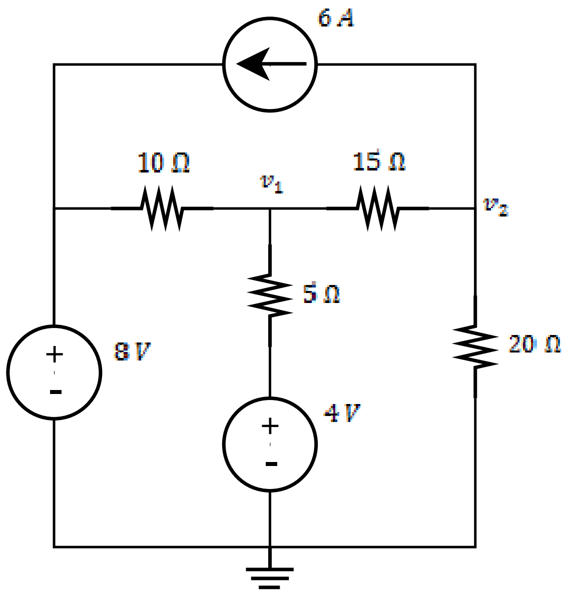
Vraag 1.6 [2]	Question 1.6 [2]
Bepaal V_{ab} en V_{bc}.	Determine V_{ab} and V_{bc}.



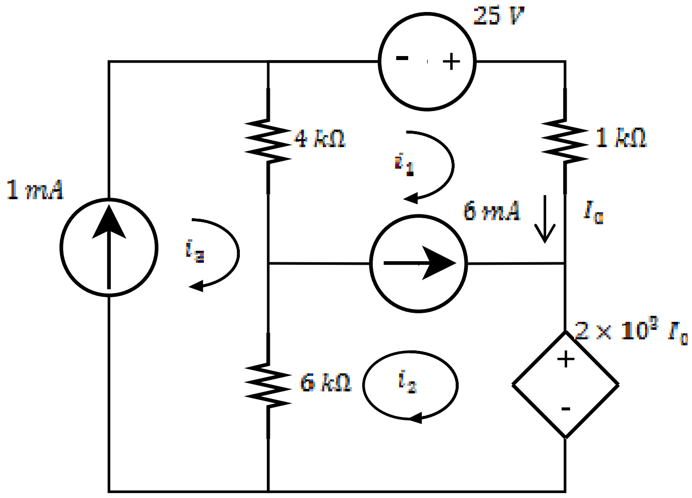
Vraag 1.7 [4] Bepaal die ekwivalente weerstand van onderstaande twee kringbane, ten opsigte van terminale *a* en *b*.	**Question 1.7 [4]** Determine the equivalent resistance of the two circuits below, with respect to terminals *a* and *b*.
Kringbaan / *Circuit* 1 	**Kringbaan / *Circuit* 2** 

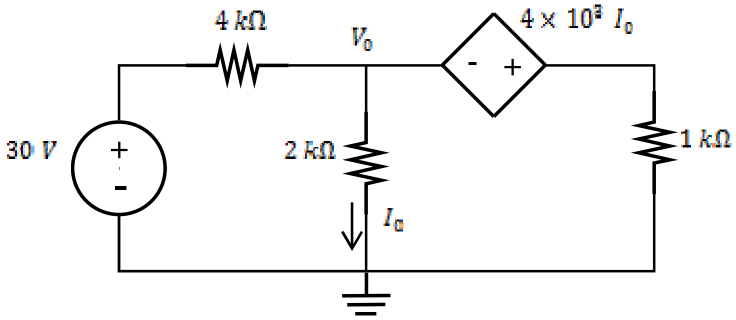
VRAAG/QUESTION 2 [32]

Vraag 2.1 [4] Gebruik node-analise om twee vergelykings van die volgende vorm op te stel: $av_1 + bv_2 = 48\ V$ $cv_1 + dv_2 = -360\ V$	**Question 2.1 [4]** Use nodal analysis to set up two equations of the following form: $av_1 + bv_2 = 48\ V$ $cv_1 + dv_2 = -360\ V$

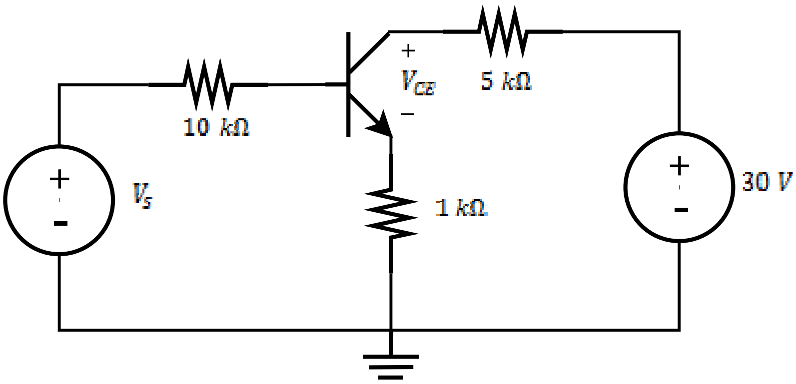


Vraag 2.2 [4] Gebruik lusanalise om twee vergelykings van die volgende vorm op te stel: $ei_1 + fi_2 = 35\ V$ $gi_1 + hi_2 = -6\ mA$	**Question 2.2 [4]** Use mesh analysis to set up two equations of the following form: $ei_1 + fi_2 = 35\ V$ $gi_1 + hi_2 = -6\ mA$

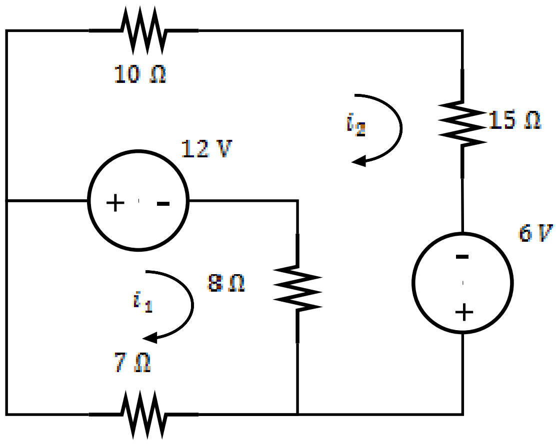


Vraag 2.3 [2] Bepaal die node spanning V_Q.	**Question 2.3 [2]** Determine the nodal voltage V_Q.
	

Vraag 2.4 [4]	**Question 2.4 [4]**
Gegee: $\beta = 100$; $V_{BE} = 0.7$ V. a) Indien $V_{CE} = 10$ V, bepaal I_C. b) Indien $V_S = 13$ V, bepaal I_B.	Given: $\beta = 100$; $V_{BE} = 0.7$ V. a) If $V_{CE} = 10$ V, determine I_C. b) If $V_S = 13$ V, determine I_B.

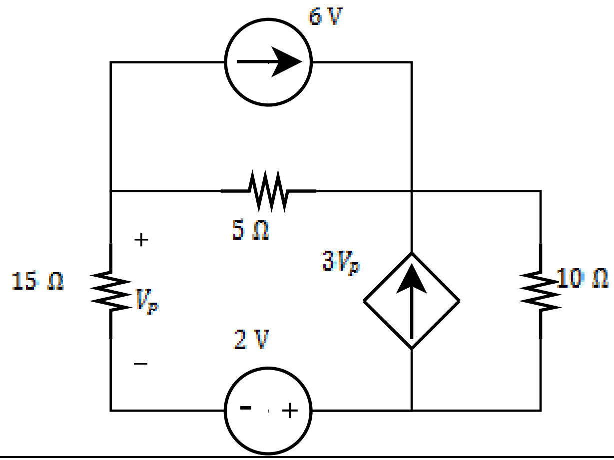


Vraag 2.5 [2]	Question 2.5 [2]
Stel `n matriksvergelyking d.m.v. inspeksie op in terme van die lusstrome.	Use inspection to set up a matrix equation in terms of the mesh currents.

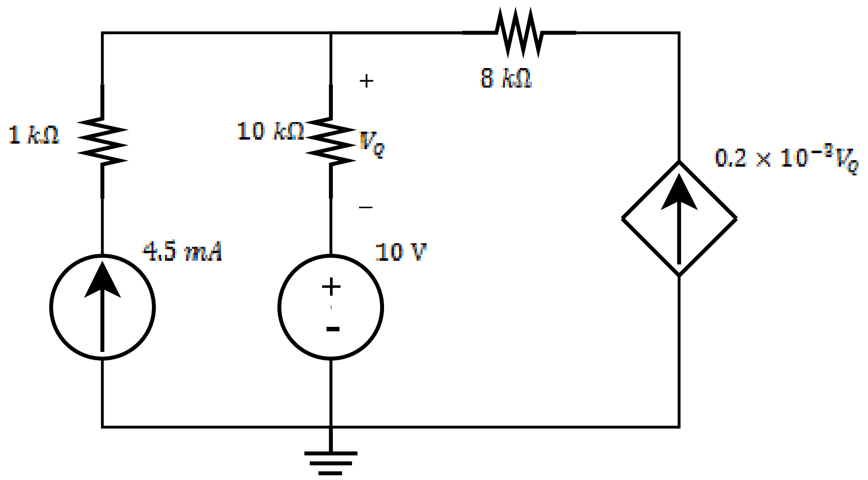


Vraag 2.6 [2]	Question 2.6 [2]
Bepaal k, indien $V_1 = 0.5$ V en $V_2 = 1$ V. $$\begin{bmatrix} 2 & -1 & 0 \\ -1 & k & -6 \\ 0 & -6 & 4 \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \\ V_3 \end{bmatrix} = \begin{bmatrix} 0 \\ -4.5 \\ 0 \end{bmatrix}$$	Determine k, if $V_1 = 0.5$ V and $V_2 = 1$ V. $$\begin{bmatrix} 2 & -1 & 0 \\ -1 & k & -6 \\ 0 & -6 & 4 \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \\ V_3 \end{bmatrix} = \begin{bmatrix} 0 \\ -4.5 \\ 0 \end{bmatrix}$$

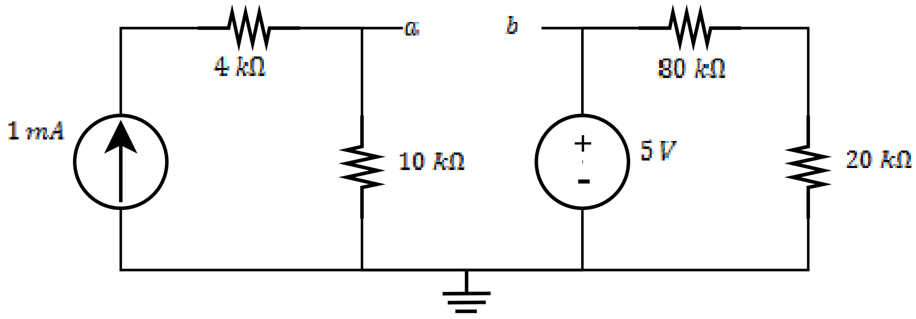
Vraag 2.7 [2]	Question 2.7 [2]
Maak gebruik van brontransformasie en bepaal V_P.	Use source transformation to determine V_P.



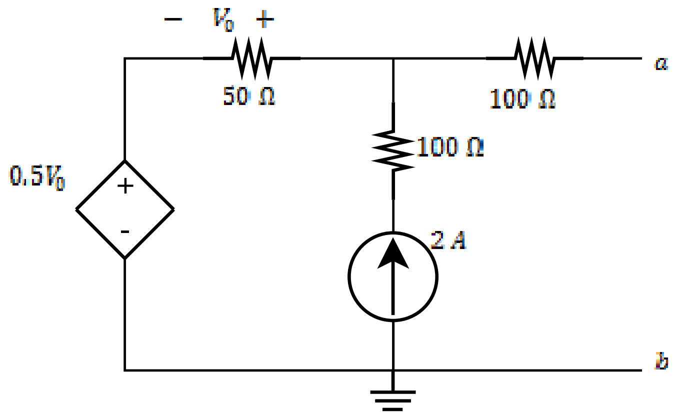
Vraag 2.8 [2]	Question 2.8 [2]
Bepaal die bydrae tot die spanning V_Q slegs as gevolg van die stroombron.	Determine the contribution of only the current source to the node voltage V_Q.



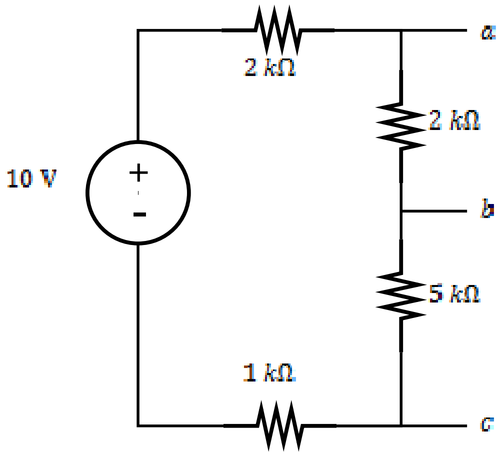
Vraag 2.9 [4]	Question 2.9 [4]
Bepaal die Thevenin ekwivalente stroombaan ten opsigte van terminale *a*-*b*.	Determine the Thevenin equivalent circuit with respect to terminals *a*-*b*.



Vraag 2.10 [4] Bepaal die Norton ekwivalente stroombaan ten opsigte van terminale *a-b*.	**Question 2.10 [4]** Determine the Norton equivalent circuit with respect to terminals *a-b*.

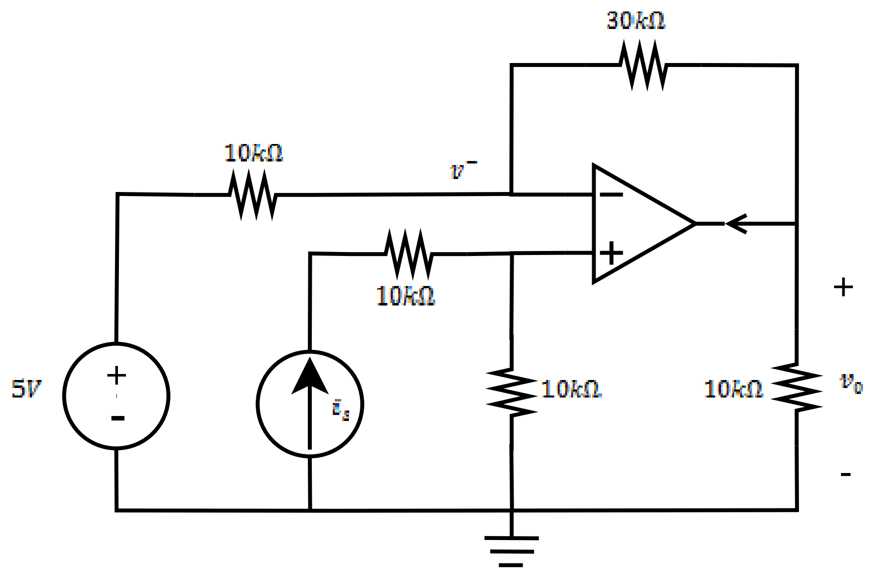


Vraag 2.11 [2]	Question 2.11 [2]
Wat is die maksimum drywing wat gelewer kan word aan \`n las tussen terminale *a* en *c*?	What is the maximum power that can be delivered to a load connected between terminals *a* and *c*?

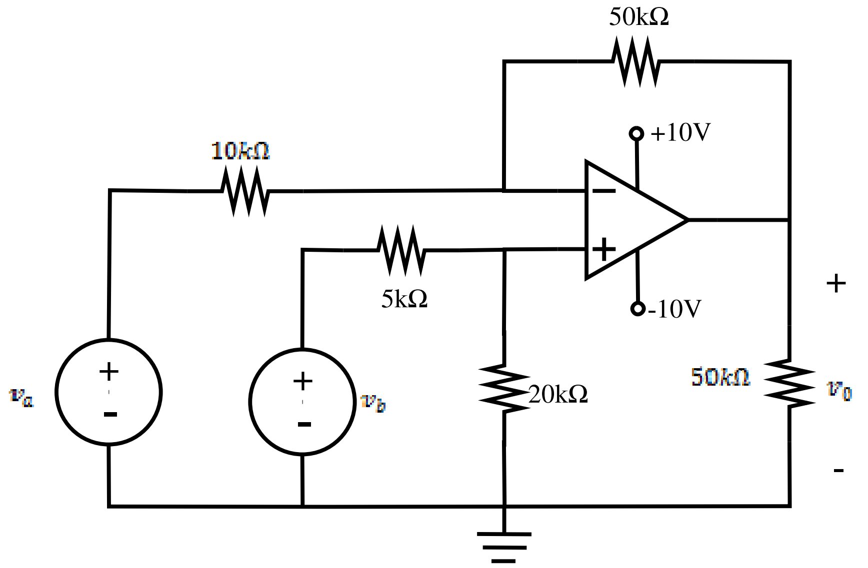


VRAAG/QUESTION 3 [18]

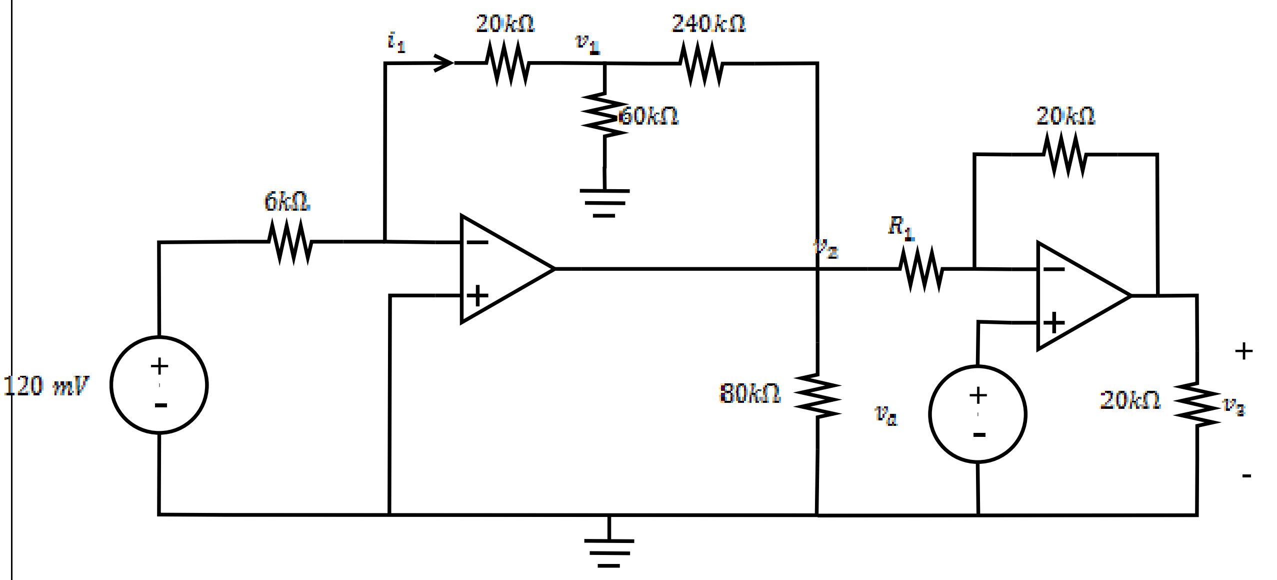
Vraag 3.1 [3]	**Question 3.1 [3]**
a) Bepaal v^- en v_0 indien $i_s = 0.1$ mA.	a) Determine v^- and v_0 if $i_s = 0.1$ mA.
b) Bepaal v_0 indien $i_s = 0$.	b) Determine v_0 if $i_s = 0$.



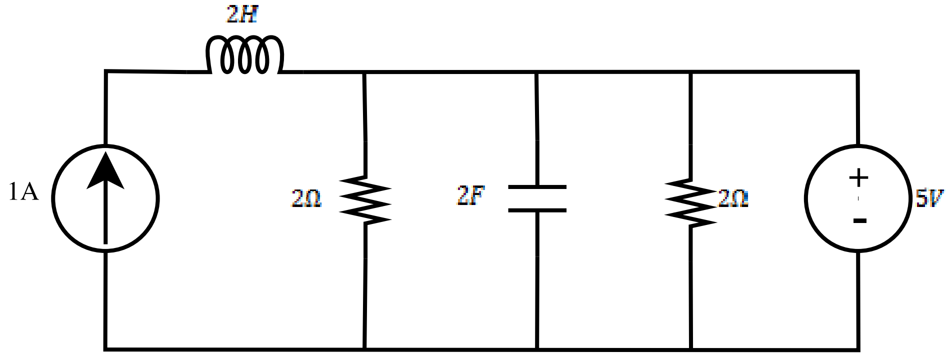
Vraag 3.2 [4]	Question 3.2 [4]
a) Bepaal v_0 as $v_a = 2V$ en $v_b = 2V$. b) Aanvaar $v_b = 4V$ en bepaal die waardes van v_a wat sal verseker dat die baan in sy lineêre gebied funksioneer.	a) Determine v_0 if $v_a = 2V$ and $v_b = 2V$. b) Assume $v_b = 4V$ and determine which range of values for v_a will result in linear operation of the circuit.



Vraag 3.3 [5]	**Question 3.3 [5]**
a) Bereken i_1, v_1 en v_2. b) Aanvaar $v_a = 0\text{V}$, en bepaal 'n waarde vir R_1 sodat $v_3/v_2 = -2$. c) Aanvaar $v_2 = -5\text{V}$, $v_a = 2\text{V}$ en $R_1 = 20\ k\Omega$, en bepaal 'n waarde vir v_3.	a) Calculate i_1, v_1 and v_2. b) Assume $v_a = 0\text{V}$, and determine a value for R_1 so that $v_3/v_2 = -2$. c) Assume $v_2 = -5\text{V}$, $v_a = 2\text{V}$ en $R_1 = 20\ k\Omega$, and determine a value for v_3.



Vraag 3.4 [2] Aanvaar die baan is in 'n stabiele toestand. Bepaal die gestoorde energie in die induktor en die kapasitor.	**Question 3.4 [2]** Assume steady state conditions in the circuit. Determine the energy stored in the inductor and the capacitor.



Vraag 3.5 [4]

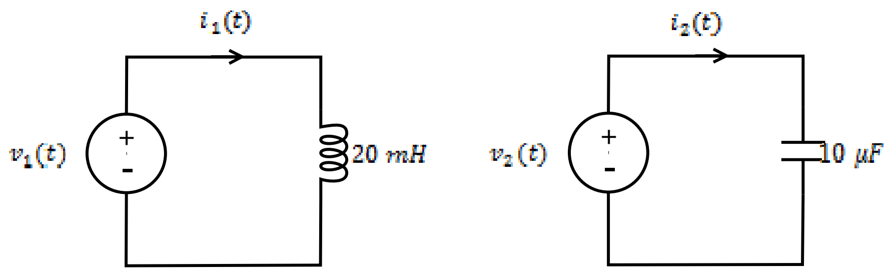
a) Beskou die stroombaan in figuur (a) en aanvaar $v_1(t) = 50 \sin 50t \; mV$ - bepaal die stroom i_1 by t = 30 ms. Aanvaar $i_1(0) = 0$.

b) Beskou die stroombaan in figuur (b) en aanvaar $v_2(t) = (3t^2 - 2) \; V$ - bepaal die stroom $i_2(t)$.

Question 3.5 [4]

a) Consider the circuit in figure (a) and assume $v(t) = 50 \sin 50t \; mV$ - determine the current i_1 at t = 30 ms. Assume $i_1(0) = 0$.

b) Consider the circuit in figure (b) and assume $v_2(t) = (3t^2 - 2) \; V$ - determine the current $i_2(t)$.



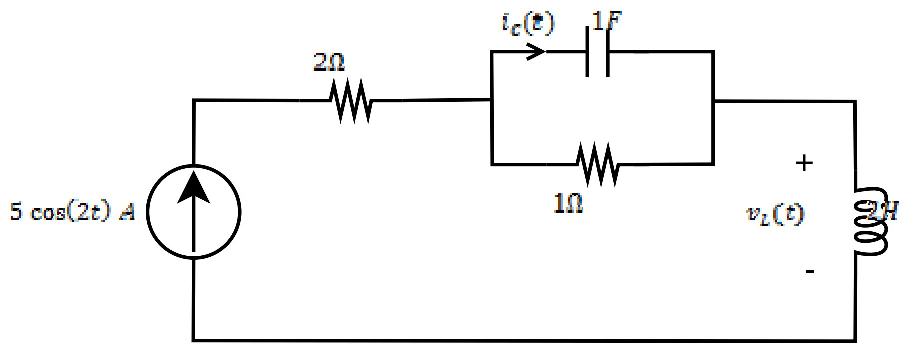
(a) (b)

VRAAG/QUESTION 4 [22]

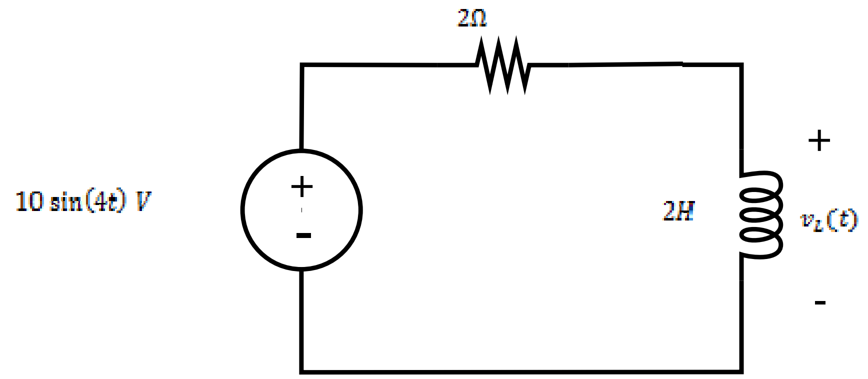
Vraag 4.1 [4]	Question 4.1 [4]
a) Skryf $i(t)$ as 'n fasor in polêre vorm: $i(t) = 21.21cos(\omega t + 90°) + 30sin(\omega t + 45°)$ A b) Watter van die volgende fasors loop voor: $\bar{V}_1 = 25\angle -130°,\ \bar{V}_2 = 10\angle 120°$ c) Bepaal die las-impedansie as $v_L(t) = 300cos(5000\pi t + 93°)$ V, en $i_L(t) = 6sin(5000\pi t + 123°)$ A. Aanvaar dat die stroom by die + van die verwysingspanning oor die las invloei. Skryf die antwoord in polêre vorm. d) 'n 40Ω weerstand en 'n $1.25\mu F$ kapasitor is in serie geskakel. 'n Spanning $v(t) = 600cos(8000t + 20°)$ V word oor die serie-kombinasie aangelê. Bereken $i(t)$. Aanvaar dat die stroom by die + van die verwysingspanning invloei.	a) Write $i(t)$ as a phasor in polar form: $i(t) = 21.21cos(\omega t + 90°) + 30sin(\omega t + 45°)$ A b) Which of the following phasors is leading: $\bar{V}_1 = 25\angle -130°,\ \bar{V}_2 = 10\angle 120°$ c) Determine the load impedance if $v_L(t) = 300cos(5000\pi t + 93°)$ V, and $i_L(t) = 6sin(5000\pi t + 123°)$ A. Assume the current is flowing into the + of the reference voltage accross the load. Write the answer in polar form. d) A 40Ω resistor and a $1.25\mu F$ kapacitor are connected in series. A voltage $v(t) = 600cos(8000t + 20°)$ V is connected accross the series combination. Determine $i(t)$. Assume the current is flowing into the + of the reference voltage.

Vraag 4.2 [4]	Question 4.2 [4]
Vraag 4.2 [4] Vereenvoudig en skryf die antwoord in reghoekige vorm: $\bar{I}_1 = \dfrac{(-3+j4)+10\angle 45^\circ}{2e^{-j20^\circ}+(6-j6)}$ Bepaal $\overline{V_2}$ in polêre vorm: $\begin{bmatrix} 2 & -j2 \\ -j2 & 3+j5 \end{bmatrix}\begin{bmatrix} \bar{V}_1 \\ \bar{V}_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 2\angle 90^\circ \end{bmatrix}$	**Question 4.2 [4]** Simplify and write the answer in rectangular form: $\bar{I}_1 = \dfrac{(-3+j4)+10\angle 45^\circ}{2e^{-j20^\circ}+(6-j6)}$ Determine $\overline{V_2}$ in polar form: $\begin{bmatrix} 2 & -j2 \\ -j2 & 3+j5 \end{bmatrix}\begin{bmatrix} \bar{V}_1 \\ \bar{V}_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 2\angle 90^\circ \end{bmatrix}$

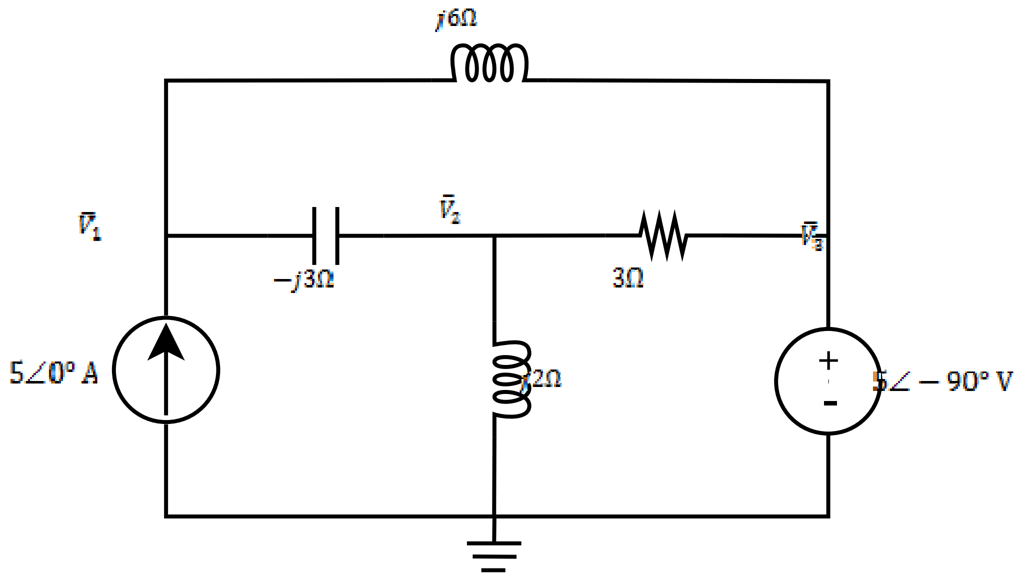
Vraag 4.3 [4]	Question 4.3 [4]
a) Bepaal die totale impedansie $\bar{Z}_{in}$ wat die bron sien. Skryf die antwoord in reghoekige vorm. b) Bereken $i_C(t)$.	a) Determine the total impedance $\bar{Z}_{in}$ seen by the source. Write the answer in rectangular form. b) Determine $i_C(t)$.



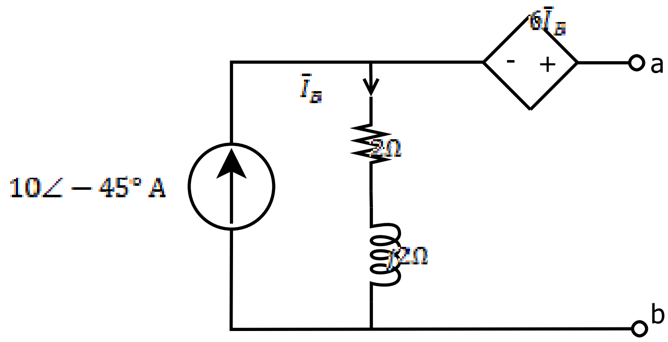
Vraag 4.4 [2]	**Question 4.4** [2]
Bereken $v_L(t)$.	Determine $v_L(t)$.



Vraag 4.5 [4]	**Question 4.5 [4]**
Gebruik node-analise om twee vergelykings van die volgende vorm op te stel: $-\bar{V}_1 + (a_1 + jb_1)\bar{V}_2 = A_2\angle\theta_2$ $(a_3 + jb_3)\bar{V}_1 + (a_4 + jb_4)\bar{V}_2 = 10\angle -90°$	Use nodal analysis to set up two equations of the following form: $-\bar{V}_1 + (a_1 + jb_1)\bar{V}_2 = A_2\angle\theta_2$ $(a_3 + jb_3)\bar{V}_1 + (a_4 + jb_4)\bar{V}_2 = 10\angle -90°$



Vraag 4.6 [4]	Question 4.6 [4]
Bepaal die Thevenin-ekwivalent tov terminale a-b. Skryf die antwoorde as fasors in polêre vorm.	Find the Thevenin equivalent wrt terminals a-b. Write the answers as phasors in polar form.



EBN111 EKSAMEN/EXAMINATION (18 JUNIE/JUNE 2013)
ANTWOORDBLAD / ANSWER SHEET

Van / *Surname:*		Studentenommer: *Student number:*								
Voorletters / *Initials:*		Tel. nr. / *Tel no.:*								

Q1		Q2		Q3		Q4		**Tot:**	**/90**

VRAAG 1 – QUESTION 1 (18)			
1.1	*a)* $i =$	*b)* $Q =$	*c)* $Cost = R$
1.2	*a)* $P_{source} =$	*b)* $V_S =$	*c)* $P_{1\&2} =$
1.3	$V_X =$	$I_X =$	
1.4	$P_Y =$	$V_Y =$	
1.5	$i_1 =$	$i_2 =$	
1.6	$V_{ab} =$	$V_{bc} =$	
1.7	$R_{ab\ (Circuit\ 1)} =$	$R_{ab\ (Circuit\ 2)} =$	

VRAAG 2 – QUESTION 2 (32)				
2.1	$a =$	$b =$	$c =$	$d =$
2.2	$e =$	$f =$	$g =$	$h =$
2.3	$V_0 =$			
2.4	*a)* $I_C =$	*b)* $I_B =$		
2.5	$\begin{bmatrix} \quad & \quad \\ \quad & \quad \end{bmatrix} \begin{bmatrix} i_1 \\ i_2 \end{bmatrix} = \begin{bmatrix} \quad \\ \quad \end{bmatrix}$			
2.6	$k =$			
2.7	$V_P =$			

2.8	$V_{Q\,(4.5\,mA)}$ =	
2.9	V_{TH} =	R_{TH} =
2.10	I_N =	R_N =
2.11	P_{max} =	

VRAAG 3 – QUESTION 3 (18)

3.1	v^- =	v_0 =	$v_{0(i_s=0)}$ =
3.2	v_0 =	$\leq v_a \leq$	
3.3	i_1 =	v_1 =	v_2 =
	R_1 =	v_3 =	
3.4	w_L =	w_C =	
3.5	$i_1(30ms)$ =	$i_2(t)$ =	

VRAAG 4 – QUESTION 4 (22)

4.1	a) $i(t)$ =	b)
	c) $\bar{Z}_L$ =	c) $i(t)$ =
4.2	$\bar{I}_1$ =	$\bar{V}_2$ =
4.3	$\bar{Z}_{in}$ =	$i_C(t)$ =
4.4	$v_L(t)$ =	
4.5	$a_1 + jb_1$ =	$A_2\angle\theta_2$ =
	$a_3 + jb_3$ =	$a_4 + jb_4$ =
4.6	$\bar{V}_{Th}$ =	$\bar{Z}_{Th}$ =